

Unrolled List – Why N linearly scales with n ?

- *Let n = total number of elements stored across the entire unrolled linked list.*
- *Let N = total number of nodes in the list.*
- *Let B = capacity of each node ($\text{MAX_ELEMENTS} = 4$)*

if $N \approx \left\lceil \frac{n}{4} \right\rceil$, then mathematically, $n \approx 4N$.

*In everyday language, we sometimes think linear means exactly equal (1: 1).
But in mathematics and computer science, linear scaling means something very specific.*

Here is exactly how, $n = 4N$ proves that N scales linearly with n .

*In mathematics, two variables scale linearly (or are directly proportional)
if one can multiply one by a fixed constant to get the other.*

*It means the equation does not have any exponents (like n^2) or
logarithms (like $\log n$).*

*In a linear relationship, the variables do not have exponents like n^2
or n^3 , nor do they have logarithms like $\log n$ or $\log_2 n$.*

*Instead, the variable is only ever raised to the **power of 1** (which is simply written as **n**). Because the equation is free of higher powers, square roots, or logarithmic curves, graphing it will always produce a perfectly straight line.*

The standard equation for a linear relationship is a straight line:

$$y = m \times x$$

In our unrolled linked list:

$$n = 4 \times N$$

*Because 4 is just a fixed constant (B), the relationship between **n** and **N** forms a perfectly straight line on a graph.*

- If $N = 1, n = 4$
- If $N = 10, n = 40$
- If $N = 100, n = 400$

Big – O notation is designed to classify algorithms by the shape of their growth curve, completely ignoring those constant multipliers. It only cares about the variables.

Because $n \approx 4N$, we can isolate N :

$$\Rightarrow N \approx \frac{1}{4}n$$

$$\Rightarrow O(N) \approx O\left(\frac{1}{4}n\right)$$

$$\Rightarrow O(N) \approx O(n)$$

The actual statement is:

$$1 \leq \text{numElements of each non – empty node} \leq B$$

Let the number of elements in node i be m_i :

$$1 \leq m_i \leq B$$

Suppose there are N non – empty nodes. The total number of elements is:

$$n = m_1 + m_2 + \cdots + m_N$$

- $m_1 = \text{number of elements in node 1.}$
- $m_2 = \text{number of elements in node 2.}$
- $m_3 = \text{number of elements in node 3.}$
- .
- .
- $m_N = \text{number of elements in node } N.$

Since every $m_i \leq B$, then:

$$\bullet \quad m_1 + m_2 + \cdots + m_N \leq B + B + B + \cdots N \text{ times}$$

$$\circ \quad n \leq NB$$

Hence,

$$N \geq \frac{n}{B}$$

or,

$$N \geq \left\lceil \frac{n}{B} \right\rceil$$

Lets say, we have:

Node 1 Elements = [1 3 4] numElements = 3	Node 2 Elements = [5 6 7 8] numElements = 4	Node 3 Elements = [9 10 11 12] numElements = 4
--	--	---

Here , $n = 3 + 4 + 4 = 11$ and $B(MAX_ELEMENTS) = 4$

$$\therefore N = \left\lceil \frac{n}{B} \right\rceil = \left\lceil \frac{11}{4} \right\rceil = \lceil 2.75 \rceil = 3$$

Now say we have :

Node 1 Elements = [1] numElements = 1	Node 2 Elements = [5] numElements = 1	Node 3 Elements = [9 10 11 12] numElements = 4
---	---	---

Here , $n = 1 + 1 + 4 = 6$ and $B(MAX_ELEMENTS) = 4$

$$\therefore N > \left\lceil \frac{n}{B} \right\rceil = \left\lceil \frac{6}{4} \right\rceil = \left\lceil \frac{3}{2} \right\rceil = \lceil 1.5 \rceil = 2$$

Hence , $N \geq \left\lceil \frac{n}{B} \right\rceil$ relation is perfect.

But because every nonempty node contains at least one element:

$$N \leq n$$

That is:

<i>Node 1</i>	<i>Node 2</i>	<i>Node 3</i>
<i>Elements = [1]</i>	<i>Elements = [5]</i>	<i>Elements = [9]</i>
<i>numElements = 1</i>	<i>numElements = 1</i>	<i>numElements = 1</i>

$$\text{Here , } N = n = 1 + 1 + 1 = 3$$

Also,

<i>Node 1</i>	<i>Node 2</i>	<i>Node 3</i>
<i>Elements = [1 2 3 4]</i>	<i>Elements = [5 6 7 8]</i>	<i>Elements = [9 10 11 12]</i>
<i>numElements = 4</i>	<i>numElements = 4</i>	<i>numElements = 4</i>

Here , $N < n = 4 + 4 + 4 = 12$

Hence , $N \leq n$,relation is perfect.

Therefore:

$$\left\lceil \frac{n}{B} \right\rceil \leq N \leq n$$

Therefore, $O(N) \subseteq O(n)$, as $N \geq \left\lceil \frac{n}{B} \right\rceil$ and $N \leq n$, and B is fixed constant.

Hence, $O(N) = O(n)$ for fixed B [This is
proved
from
above]

$\therefore$ With fixed B :

$$N = \Theta(n)$$

*$\therefore$ Scaling **N** nodes blocks with **n** number of elements is Linear.*
